

Comparing Interpretable Sequence Features and Protein Language Models for Missense Variant Classification

Georgios Ioannou (gi2100), Vedant Jagtap (vsj7589)

1. Background/Motivation

Interpreting whether a human genetic variant is likely damaging is an important question in genomics and precision medicine. Many strong tools already exist for missense variant effect prediction, so our goal is not simply to reach state-of-the-art performance. Instead, we will use ClinVar missense variants to study how much pathogenicity signal is captured by simple, interpretable sequence features, and whether pretrained protein language model embeddings improve cross-gene generalization beyond those features. As of March 30, 2026, ClinVar reports 4,235,985 unique variation records with germline classifications and variants in 17,567 protein-coding genes, making it suitable for a focused supervised learning study and a realistic gene-held-out evaluation.

2. Problem formulation

We will build binary classifiers that predict whether a human missense variant is pathogenic/likely pathogenic or benign/likely benign. Rather than aiming to outperform established pathogenicity predictors, we will compare handcrafted sequence features with ESM-2 protein language model embeddings to test the tradeoff between interpretability, computational simplicity, predictive performance, and cross-gene generalization.

3. Data & methods

Our primary dataset will be ClinVar. We plan to use NCBI's downloadable variant_summary.txt file, which represents all ClinVar variants that have a location on the genome and is released monthly. We will restrict the analysis to germline missense variants with clear benign/likely benign or pathogenic/likely pathogenic labels, and we will exclude variants with uncertain significance or conflicting classifications. To improve label reliability, we will prioritize records with stronger review statuses, such as expert-reviewed records, while assessing the tradeoff between label quality and dataset size. Because this is only a proposal, we have not yet run preprocessing. The exact number of retained missense variants will be determined after approval and reported in the final project.

For sequence context, we plan to map variants to human protein sequences from the UniProt human reference proteome (UP000005640), which should provide broad sequence coverage for this project. We will include only variants that can be mapped confidently to a reviewed human protein sequence. Our baseline model will be logistic regression or random forest trained on handcrafted sequence-based features, including the reference amino acid, alternate amino acid, BLOSUM62 substitution scores, local sequence-window composition, and normalized residue position. The model input will therefore be a missense variant and its protein-sequence context, and the output will be the probability that the variant is pathogenic. The purpose of this baseline is not only to provide a performance comparison, but also to test whether simple and interpretable biological features can explain a meaningful fraction of variant pathogenicity.

Our stronger model will use residue-level embeddings from ESM-2 (650M parameters), extracted at the mutated position of each protein sequence, and the output will again be the probability that the variant is pathogenic. We note that ESM-2 was pretrained on large protein sequence corpora that include human proteins, which could slightly inflate evaluation metrics. If feasible, we will compare our models against established pathogenicity predictors such as CADD and AlphaMissense to contextualize our results, but these tools will be treated as reference points rather than targets we expect to outperform. We will evaluate all models using AUROC, AUPRC, F1 score, and confusion matrices. Because pathogenic and benign labels may be imbalanced, we will consider class weighting

or balanced sampling during training. To reduce train-test leakage, our main split will hold out entire genes rather than randomly splitting variants from the same gene across train and test sets.

References

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